Ontario's Global Adjustment

Anatomy, Facility-Level Evidence, and the 2027–2035 Outlook

Kilambi Research

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Abstract. Ontario's Global Adjustment (GA) is the charge that settles the difference between what the province's contracted and regulated generators are owed and what the wholesale market pays them. This paper builds the charge from its sources: a monthly actual series for January 2005 to September 2026 assembled from the Independent Electricity System Operator's own publications, a contract register of 3,582 facilities, and a facility-level reconstruction of the market revenue each generator earned, priced hour by hour at the market price (the Hourly Ontario Energy Price to April 2025, delivery-point nodal prices after the 2025 market renewal). Three findings follow. First, GA is a closed three-pool system whose published components sum to the total (in every month from 2011, and within C$1.2 million in the early workbook years), and its size is set by the contract book rather than by current market conditions: over the ten years to September 2026 it collected C$94.48 billion against C$47.11 billion of wholesale commodity value, C$2.01 of GA for every C$1 of energy. Second, the facility record prices 99.03 percent of 2024 generation but contract totals are public for only part of the fleet, so the reconstruction explains 43.5 percent of 2024 GA (C$3.67 billion of C$8.43 billion); the unexplained balance is named, and its largest block, Bruce Power's unitemised top-up, is bounded at C$1.53 billion to C$2.04 billion for 2024. Third, a forecast built pool by pool for 2027 to 2035 gives Total GA of C$6.79 billion to C$11.58 billion in 2027 and C$5.38 billion to C$12.70 billion in 2035 across the scenario grid. Under current Ontario Power Generation regulated economics the total declines in every cell; under the EB-2025-0297 settlement path it rises to a 2031 peak, because settled payment amounts grow faster than the contract portfolio decays. A backtest of the same machinery over 2020 to 2024 errs by −3.1 percent in the one year with no policy change and by +23.5 percent cumulatively, with the 2021 decision to move most non-hydro renewable contract costs to the tax base accounting for most of the miss. The backtest result is the paper's central caution: GA forecasts fail on perimeter decisions, not on prices.

Keywords: Global Adjustment; Ontario electricity market; contracts for difference; regulated tariffs; electricity cost allocation; market renewal

JEL codes: L94; L51; Q48

Non-technical summary. Most of the electricity generated in Ontario is not sold at the market price. It is sold under contracts, or under regulated rates, that promise the generator a fixed amount per unit of output. When the market price is below that amount, somebody has to pay the difference. In Ontario that payment is collected through a line on the electricity bill called the Global Adjustment. This paper measures the charge directly: how large it has been since 2005, which generators it pays, and how much of it can be accounted for facility by facility from public records. It then forecasts the charge to 2035 under a range of market prices and under the two regulatory paths now open for Ontario Power Generation. The main results are that the charge has recently run at about twice the value of the energy itself, that just under half of it can be traced to individual facilities from public data, and that its future path depends less on electricity prices than on two things governments and regulators control: which costs are placed inside the charge, and the rates set for the province's largest generator.

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1 Introduction

Ontario buys most of its electricity twice. The wholesale market pays generators the Hourly Ontario Energy Price (HOEP), and since the market renewal of 2025 a set of locational prices, for each megawatt-hour they produce. Separately, most generators hold a contract or a regulated rate that promises a fixed total amount per megawatt-hour. The Global Adjustment (GA) is the monthly charge that settles the difference between the two payments across the whole system. When market prices fall, GA rises by close to the same amount per unit of contracted output; when market prices spike, GA compresses and can turn negative. The charge is therefore not a tax on electricity and not

a market outcome. It is the residual of a contracting decision, and its size in any month is determined mainly by decisions taken years earlier about what to contract, at what price, and for how long.

That makes GA a measurement problem before it is a policy problem. The Independent Electricity System Operator (IESO) publishes GA monthly, split into three components, and publishes the contract register and the output and price records from which a facility-level account can be built. Public discussion of GA has mostly worked from the published totals. This paper works from the facility record up: it prices every generator's output at the market price it actually faced, hour by hour, and compares the result with what the generator was owed. The gap between the two, summed over facilities, should reproduce GA. Where it does not, the shortfall identifies exactly which parts of the charge the public record cannot itemise.

Three findings organise the paper.

First, GA is a closed three-pool system, and its scale relative to the market is large and persistent. The published components, GA-OPA (the contract portfolio), GA-OPG (Ontario Power Generation's prescribed fleets) and GA-OEFC-NUG (legacy non-utility generator contracts), sum to Total GA in all 132 months of the modern workbook era, verified to the cent in our assembly (Observed, A). Total GA rose from a credit of C$1.15 billion in 2005 to a peak of C$13.96 billion in 2020, and over the ten years to September 2026 it collected C$94.48 billion against C$47.11 billion of wholesale commodity value: C$2.01 of GA for every C$1 of energy (Reconstructed, A). The Class B rate that most customers face peaked at C$118.19/MWh in 2020 and fell to C$42.59/MWh in 2025, a path driven less by contract additions than by two perimeter decisions, the 2017 Fair Hydro reductions and the 2021 move of most non-hydro renewable contract costs to the tax base.

Second, the facility record is wide on prices and narrow on contract totals. The derived rates table built for this paper prices the output of 115 facilities, covering 99.03 percent of 2024 generation (Reconstructed, A). Contract totals, however, are public for only 37 of those facilities, because the province's pooled contract programs disclose prices at program level or not at all. The reconstruction therefore explains 43.5 percent of 2024 GA (C$3.67 billion of C$8.43 billion) and 37.6 percent over 2015 to 2024 (Reconstructed, A). The unexplained C$4.76 billion in 2024 is not a mystery residual: it sits inside GA-OPA, and its largest single block is Bruce Power's top-up, which is settled inside the pool and never itemised. Held documents bound that top-up at C$1.53 billion to C$2.04 billion for 2024, 28 percent of GA-OPA at the low bound. The calibration evidence runs the other way where totals are public: the same machinery reproduces Ontario Power Generation's stated blended rate for January to April 2025 within 0.81 percent.

Third, the charge's future path depends on a regulatory branch, and the two branches point in opposite directions. We forecast GA pool by pool for 2027 to 2035 across three market-price scenarios and two price bases for Bruce. Under branch A, which holds Ontario Power Generation's 2024 implied regulated economics flat, Total GA declines from 2027 to 2035 in every cell of the grid, from a 2027 range of C$6.79 billion to C$11.18 billion to a 2035 range of C$5.38 billion to C$9.77 billion, because the contract portfolio decays through dated expiries while demand grows 25.0 percent (Modelled, B). Under branch B, which substitutes the settlement path in Ontario Energy Board case EB-2025-0297 (settled weighted average payment amount of C$102.79/MWh in 2027 rising to C$124.46/MWh in 2031, against C$77.75/MWh approved for 2026), Total GA rises in every cell and peaks around 2031. The Class B rate falls in every cell under both branches, because the denominator grows faster than either numerator. A backtest disciplines how much of this to believe: run as of end-2019 with actual prices fed in, the machinery misses 2020, a year with no policy change, by −3.1 percent, and misses 2021 to 2024 by 29 to 48 percent per year, with the 2021 tax-base shift accounting for most of the error. Perimeter decisions are the forecast risk; prices are second.

The paper contributes three things. It assembles the first continuous monthly GA series from 2005 with a verified component identity and a named provenance for the widely quoted ten-year total. It builds the facility-level market-leg table for the Ontario contract register and states its coverage honestly, including the correction of an earlier double count of Bruce capacity and the withdrawal of an unreproducible Bruce price figure that had entered internal diagnostics. It then carries the same accounting into a forecast whose assumptions are labelled line by line and whose branch structure reflects the actual regulatory state of play in November 2026: a comprehensive settlement in EB-2025-0297 that has been presented to the Ontario Energy Board and not yet decided.

The rest of the paper proceeds as follows. Section 2 sets out how GA works institutionally: the three pools, the two customer classes, and the dated policy changes that move costs into and out of the charge. Section 3 places the work in the literature on contract design and cost pass-through. Section 4 describes the datasets, and Section 5 the computations: the derived market legs, the calibration, the top-up identity, and the forecast machinery. Sections 6 to 9 report results in four parts: the anatomy of the actual charge, the facility-level evidence and reconciliation, the roll-off schedule, and the forecast with its backtest. Section 10 discusses what the results imply for reading GA as a policy variable. Section 11 states the limitations, including what evidence would change the conclusions. Section 12 concludes.

2 Institutional background

2.1 What the Global Adjustment settles

Ontario's wholesale market, operated by the IESO, pays generators a uniform provincial price in the pre-renewal design (HOEP) for energy delivered. Most supply, however, was procured outside the spot market. Successive Ontario governments contracted generation through the Ontario Power Authority (OPA, merged into the IESO in 2015) under programs that pay a fixed or formula price per megawatt-hour: feed-in tariffs (FIT) for renewable projects, Renewable Energy Supply (RES) contracts, the Large Renewable Procurement (LRP) round, Clean Energy Supply and Advanced Clean Energy Supply contracts for gas plants, and heritage contracts for waterpower. Ontario Power Generation (OPG), the provincial generator, is paid regulated payment amounts set by the Ontario Energy Board (OEB) for its prescribed nuclear and hydroelectric fleets. Bruce Power, the private operator of the Bruce nuclear station, is paid under the Bruce Power Refurbishment Implementation Agreement (BPRIA) at a base price of C$65.73/MWh in 2016 terms, indexed and stepped for the refurbishment program.

For each of these arrangements the generator receives the market price from the market and a top-up (or pays back a surplus) so that its total equals the contracted or regulated amount. GA is the system's monthly sum of those top-ups, recovered from customers. It is, in the vocabulary of the wider literature, a gap charge on a portfolio dominated by two-way contracts for difference and regulated tariffs; a companion paper in this series (Kilambi Research, KR-WP-2026-37) sets Ontario alongside the equivalent mechanisms in other large electricity systems.

2.2 The three pools

The IESO publishes GA in three components, and the decomposition is closed: the components sum to the total in every month from 2011, and to within C$1.2 million in 12 months of 2008 to 2010, where the early workbook published rounded totals.

GA-OPA is the contract portfolio administered by the IESO as OPA's successor: FIT, RES, LRP, CES and ACES contracts, heritage waterpower contracts, and two items that are never itemised in the publications, Bruce Power's top-up and the former OPA's conservation (CDM) spending. It is the largest pool in most years: C$9.22 billion of the C$13.96 billion total in 2020, and C$5.45 billion of C$8.43 billion in 2024 (Observed, A).

GA-OPG is the top-up on OPG's prescribed fleets, equal by construction to the fleets' regulated revenue requirement less their market revenue. It ranged from C$1.66 billion to C$4.64 billion per year over 2015 to 2025 in the assembled series.

GA-OEFC-NUG is the Ontario Electricity Financial Corporation's legacy non-utility generator contracts, signed in the 1990s. It decayed from C$0.72 billion in 2015 to C$0.03 billion in 2024 as the contracts expired, and it is the worked example in the data of what a completed roll-off looks like.

Two charges often confused with GA are not in it. Hourly uplift, which recovers operating reserve and related costs, is charged through the market's hourly uplift mechanism. The Rural or Remote Rate Protection and the Ontario Electricity Support Program are set by the OEB and recovered through distribution rates or provincial funding. Neither appears in the GA decomposition in any held publication (Section 4).

2.3 Who pays: Class A and Class B

GA is allocated between two customer classes. Class A, large customers whose peak demand qualifies them for the Industrial Conservation Initiative, pays a share of GA set by its contribution to the system's top peak hours (the peak demand factor), not by its energy consumption. Class B, everyone else, pays the residual, divided by Class B consumption to give the Class B rate in C$/MWh, the number most Ontario customers experience as the GA line.

The allocation has shifted over time. Class A paid 11.0 percent of GA dollars in 2015 and 16.7 percent in 2024, while consuming 18.8 percent of energy in 2015 and 30.7 percent in 2024 (Reconstructed, A). Class B consumption ran between 97.8 and 114.0 TWh per year over 2015 to 2025. Because Class A's share is set by peak contribution, GA's incidence is a design choice that can be, and has been, adjusted without changing the total.

2.4 Dated events that move the perimeter

GA's history is punctuated by decisions that changed what the charge contains. Four matter for reading the series and for the forecast.

The Fair Hydro Plan (2017) reduced bills by refinancing a portion of GA costs over a longer horizon; the Financial Accountability Office of Ontario estimated the plan's cost and accounting treatment at the time (Financial Accountability Office of Ontario, 2017). The published Class B rates for July to November 2017 and December 2018 sit below the rates implied by the workbook amounts and consumption, the fingerprint of that period's adjustments in the series we assembled.

The 2020 deferral capped the Class B rate at C$115.00/MWh for April to June 2020 during the pandemic; the deferred amounts were recovered through 2021 at published recovery rates. At annual grain the deferral nets out; at monthly grain it is plainly visible.

The tax-base shift of January 2021 is the largest perimeter change in the series. From that date the published GA-OPA excludes about 85 percent of non-hydro renewable contract costs, which the province moved to the tax base. GA-OPA fell from C$9.22 billion in 2020 to C$5.15 billion in 2021 while the underlying contracts continued unchanged. GA before and after 2021 is not on one basis,

and any comparison across that date must say so.

The market renewal of 1 May 2025 replaced the uniform HOEP with locational marginal prices. GA's settlement logic is unchanged, but the market leg against which top-ups are computed is now each generator's delivery-point price, and the renewed market's first-year true-ups have been wide: actual GA for January and February 2026 was a credit of C$493.9 million and C$344.6 million respectively (Class B rates of −$40.62/MWh and −$32.81/MWh), after positive estimates (Observed, A).

2.5 The open regulatory case: EB-2025-0297

OPG's prescribed-fleet payment amounts are set in five-year rate periods. The current period ends 31 December 2026. In case EB-2025-0297, OPG and DNNP LP applied in December 2025 for payment amounts for 2027 to 2031 covering the regulated hydroelectric fleet, the nuclear fleet, and the Darlington New Nuclear Project. A settlement conference produced a comprehensive Settlement Proposal, filed 5 October 2026 with a draft Payment Amounts Order, which was presented to the OEB on 30 October 2026. No Decision and Order had been issued at the date of this paper's data build. The settlement's weighted average payment amount is C$102.79/MWh for 2027, rising to C$124.46/MWh in 2031, against C$77.75/MWh approved for 2026 (Appendix A, Table 9). The settlement is comprehensive on all issues except the clearance of a C$170.2 million debit balance in the Capacity Refurbishment Variance Account for the Darlington Refurbishment Project, which proceeds separately, and the settled amounts remain subject to an update for the OEB's prevailing return on equity at the date of the final order. The forecast in Section 9 carries both the current-economics path and the settlement path as separate branches: by 2031 the settlement adds C$2.98 billion to Total GA on the central market path, second only to the market-price scenario itself among the forecast's drivers, and it reverses the direction of the total.

3 Related literature

This paper is a measurement exercise on a specific charge, and three literatures bear on it.

The first is the design literature on publicly backed generation contracts. Two-way contracts for difference, under which a generator is topped up when the market price is below a strike price and pays back when it is above, convert a volatile market revenue into a fixed one and move the price risk to whoever funds the difference. Joskow (2008) sets out the wider lesson from electricity liberalisation that contract structure, not the spot market alone, determines what investors earn and what consumers pay. Ontario's GA is the accounting residue of exactly that choice, made contract by contract over two decades: the charge is large because the contracted share of supply is large and the contract prices were set, for the early renewable fleet, well above subsequent market prices. The gap-charge form is not unique to Ontario; KR-WP-2026-37 (Kilambi Research, 2026) documents levies with the same structure in the United Kingdom, Germany, Spain and Japan, and works a stylised example showing that a C$20/MWh fall in the market price moves average consumer cost by only C$6.00/MWh in year one when the contracted share is high. The present paper is the facility-level companion for the reference case in that study.

The second literature measures the market value of generation by technology. Hirth (2013) shows that wind's capture price falls relative to the average market price as wind penetration rises, because wind output is correlated across projects and concentrates in lower-priced hours. Our derived legs measure the same object directly for Ontario: the wind fleet's output-weighted market leg ran below the covered-fleet leg in every year from 2010 to 2024, by 6.2 percent in the closest year (2019) and

Table 1: The datasets built for this paper

Dataset	Rows	Span	Coverage statement
GA actual series (monthly)	261	2005-01 to 2026-09	Total GA and its three components (components from 2008); class amounts and consumption from 2015
Contract register	3,582	contracts to 2067	32,428.8 MW stated capacity
Derived market rates	16,741	2010 to 2026	115 facilities with market legs; 37 carry a contract total
Fuel price panel (monthly)	1,265	2010-01 to 2026-11	189 months; no rows for the 2025-05 to 2026-06 price gap
Roll-off schedule	168	end years 2026 to 2046+	3,553 of 3,582 register rows carry a term-end year
GA forecast grid	108	2027 to 2035	3 leg scenarios × 2 OPG branches × 2 Bruce bases

Notes: Row counts are the files' own counts, verified at build. Derived market legs price output at HOEP to April 2025 and at delivery-point nodal prices from 23 July 2026; no prices are held for the interval between, and no series in this paper crosses that gap. *Source:* Kilambi Research IESO data estate (Appendix A, Table A3, lists vault keys and hashes).

25.6 percent in the widest (2020) (Reconstructed, A). The discount is the reason a fixed-price wind contract needs a larger top-up per megawatt-hour than the system average price gap suggests, and it is a structural feature of the charge rather than a forecasting error by any party.

The third literature is the Ontario policy record itself. The Auditor General of Ontario's 2015 report on electricity power system planning criticised the cost of the contract portfolio and the size of GA (Auditor General of Ontario, 2015). The Financial Accountability Office assessed the Fair Hydro Plan's fiscal effect in 2017 (Financial Accountability Office of Ontario, 2017). The IESO and the OEB publish the primary material: GA workbooks and monthly notices (Independent Electricity System Operator, 2026b), the Annual Planning Outlook's demand forecast (Independent Electricity System Operator, 2026a), and the EB-2025-0297 record including the Settlement Proposal (Ontario Power Generation Inc. and DNNP LP, 2026). What that record has not contained is a facility-level reconciliation: published GA is a pool total, and neither the IESO nor any study we can draw on itemises the pool against the register of contracts and the market revenue each facility earned. Sections 6 and 7 supply that reconciliation and quantify how much of the pool the public record can and cannot support.

4 Data

All data are IESO or OEB publications, held in the Kilambi Research IESO data estate. Appendix A, Table 10, lists each file with its vault key and hash. All dollar figures are Canadian dollars. Table 1 inventories the datasets built for this paper.

4.1 The GA actual series

We assembled a monthly series of Total GA, its three components, and Class A and Class B amounts and consumption for January 2005 to September 2026, 261 months, from three held sources. For 2005 to 2014 we used the IESO's Global Adjustment Values workbook, which carries totals throughout, components from 2008, and class dollar totals from 2011. For 2015 to 2025 we used the IESO's Global Adjustment Components workbook, which carries components, class amounts and class consumption monthly; in that source the three components sum to the published Total GA in all 132 months, a check our assembly reproduces to the cent. For January to September 2026 we used the IESO's monthly Class B rate notices: the notices' actual dollar field is Total GA, not the Class B portion, which we verified on all three overlap months with the workbook (October to December 2025, matching to the cent or to the workbook's rounding). An independent estate compilation from IESO settlement publications matches the assembled annual totals to C$0.1 million in every year from 2015 to 2025. October 2026 (second estimate only) and November 2026 (first-estimate rate only) are held but are not actuals and are excluded.

Two definitional points matter. The Class B rate in the series is implied, Class B amount over Class B consumption, so that it is internally consistent with the held amounts; it matches the IESO published actual rate to the cent in most months, and the months where it differs by more than C$1/MWh cluster in the Fair Hydro windows and in April and May 2020, when the published rate is exactly C$115.00, the government cap. And the IESO file that appears to offer a by-components split for 2005 to 2014 (GA-by-Components.xlsx) is in fact an IESO 404 error page in the vault; no finer pre-2015 split is held, and none is used.

4.2 The contract register and the derived rates

The contract register is built from the IESO's Active Contracted Generation List and carrying documents: 3,582 rows over 32,428.8 MW of stated capacity, each row carrying, where the public record supports it, a contract price and an exactness flag (EXACT, TARIFF, AGGREGATE-ONLY, NOT PUBLIC), and term start and end dates. Prices are public at facility level for the tariff programs and for Bruce's 2016 base price; the pooled programs (CES, ACES, GEIA, RES, LRP and the capacity procurements) are AGGREGATE-ONLY by the structure of their disclosure.

The derived market rates table, built for this program and described in Section 5, prices each register facility's output at the market: 16,741 monthly and annual rows for 115 facilities. Its coverage needs care in both directions. On nameplate, the table covers 129 distinct register rows totalling 23,666.9 MW, 73.0 percent of the register. An earlier build reported 30,215.9 MW and 93.2 percent; that figure double-counted Bruce Power, whose 6,549.0 MW appears both in two station-aggregate rows used for 2010 to 2014 reporting and in the eight unit rows. We use the corrected figure throughout. On energy, coverage is far higher: the derived facilities produced 155.36 TWh of Ontario's 156.88 TWh in 2024, 99.03 percent, and 12.42 TWh of 12.67 TWh in October 2026, 97.99 percent. The uncovered nameplate is mostly capacity that has not yet generated (4,216.2 MW of LT 1, LT 2 and ELT 1 procurement winners not yet in service) or that generates behind the meter in aggregate-only form (2,547.8 MW of distribution-connected FIT and RESOP projects). Neither gap is fillable at facility level from held data; both are named again in Section 11 because they bound what the fuel panel can say.

Contract totals exist for 37 of the 115 priced facilities: 36 tariff facilities (the HCI waterpower stations at C$69.00/MWh, the FIT wind farms with output feeds, and one solar reporting aggregate) plus a combined OPG prescribed-fleets row whose total is implied by GA-OPG itself. Bruce carries market

legs only. An earlier internal diagnostic used a Bruce total of C$83.12/MWh for 2025; that figure is not reproducible from held files (the register carries the 2016 base of C$65.73/MWh, and the IESO backgrounder on the agreement describes about C$83/MWh as the contemporaneous average commodity cost the new price sat below), and the dollar amounts built on it are withdrawn. Section 7.3 uses only the documented bounds, C$65.73/MWh and a stated life average of about C$77/MWh.

4.3 Prices

Pre-renewal prices are HOEP from the IESO's annual realtime market price files, hourly means of the 5-minute values, continuous from January 2010 to April 2025 (134,376 hours). We verified the series against the IESO's published monthly averages in all 184 months; the worst difference on the 2010 to 2014 extension is C$0.0053/MWh. Post-renewal prices are the parsed realtime price table's energy product at each generator's delivery-point nodes: 1,065 locations, no provincial reference price. A three-day October 2026 sample of 219,456 interval-location keys showed zero disagreement across published vintages. No prices are held for 1 May 2025 to 22 July 2026 (the first fifteen months of the renewed market), and 1 August 2026 is also missing. Market legs therefore exist for the HOEP era to April 2025 and for the renewed market from 23 July 2026, and nothing in this paper interpolates across the gap.

4.4 Demand and the regulatory record

The demand denominator is the IESO's 2026 Annual Planning Outlook, Demand Forecast Module, extracted in full for this program: Reference-scenario net demand of 156.31 TWh in 2027 rising to 195.33 TWh in 2035. The workbook is demand-only; it carries no generation-by-fuel outlook for grid-connected supply, a fact that forces the forecast's constant-mix assumption (Section 5.4). Its Table 5 does carry embedded generation by fuel, which falls from 7.34 TWh in 2027 to 3.27 TWh in 2035. The regulatory record for branch B is the EB-2025-0297 Settlement Proposal and its draft Payment Amounts Order (Appendix A, Table 9).

5 Methodology

5.1 The derived market leg

For each facility f and period t (month or year), the market leg is the output-weighted average market price its generation earned:

$$\text{leg}_{f,t} = \frac{\sum_{h \in t} p_{h(f)}\, q_{f,h}}{\sum_{h \in t} q_{f,h}}, \tag{1}$$

where $q_{f,h}$ is facility output in hour h from the IESO generation capability reports (deduplicated to the latest published vintage per generator-hour, with the reports' provincial pseudo-generator excluded), and $p_{h(f)}$ is HOEP in the pre-renewal era or the equal-weighted mean of the facility's delivery-point nodal prices in the renewed era. Where output is published for a station aggregate in one era and for units in another, mapping priority rules prevent any hour being counted twice. Storage facilities price their discharge series only.

Where a facility's contract total T_f is known, its contract top-up is $T_f - \text{leg}_{f,t}$ and its GA contribution is that top-up times output. Register totals are base schedule rates: escalation and indexation clauses are recorded in the register as text and are not arithmetically applied, because no per-facility index

series is held. Every result that uses a total inherits that convention, and we flag it where it binds (Section 11).

5.2 Calibration

The construction is gated on external checks. For the OPG prescribed fleets, GA-OPG is published, so the fleets' implied total can be computed as (market revenue + GA-OPG) / output and compared with OPG's stated blended rate. For January to April 2025 the derived table gives an output-weighted implied total of C$81.56/MWh against the stated C$80.91/MWh, a difference of +0.81 percent (Reconstructed, A). We recomputed that figure from the held files for this paper: the combined fleets' monthly implied totals in the table (C$82.46, C$83.48, C$82.28 and C$78.13/MWh for January to April 2025) are each the month's market leg plus that month's published GA-OPG divided by the month's output, and their output-weighted mean is the C$81.56/MWh above. Facility-level values carry their own checks. The table's build documentation records three hand spot checks, computed from the raw monthly capability and price files through an independent code path; we re-performed all three against the held table for this paper and confirmed each one exactly: Goreway Station in June 2023, a market leg of C$35.38/MWh; Clergue GS in September 2019, a leg of C$13.37/MWh against a contract total of C$69.00/MWh; and Wolfe Island Wind in June 2023, a leg of C$25.74/MWh (Reconstructed, A). The table's arithmetic binds internally as well: on all 5,870 facility-period rows that carry a contract total, total less leg less top-up is zero to within C$0.0001/MWh. The table's daily standing updater reproduces closed periods byte-identically, which is the sense in which the numbers in this paper are stable: regenerating the derived table from the same inputs returns the same values.

5.3 The top-up identity and the reconciliation

The reconciliation tests the identity

$$\text{GA} \approx \sum_{f \in \text{priced, totals known}} (T_f - \text{leg}_f)\, Q_f \;+\; \text{GA-OPG} \;+\; \text{unmodelled pools}, \tag{2}$$

where the OPG term enters through the combined prescribed-fleets row. Two parts of the sum are computable from the facility record: the OPG row, whose top-up is GA-OPG by construction (agreement there is a build check on the output and price machinery, not independent validation), and the schedule-priced subset of HCI hydro, FIT wind and one solar aggregate. Everything else is unmodelled by design, and the reconciliation's content is the size and composition of the remainder, stated in Section 7.3.

5.4 The forecast machinery

The forecast is built pool by pool at annual grain for 2027 to 2035, calibrated on 2024, the latest year with a complete held decomposition and full-year derived legs.

GA-OPG, branch A, holds the 2024 calibration: prescribed-fleet output of 67.90 TWh times an implied total of C$76.48/MWh less the scenario market leg (C$32.98/MWh in 2024). Branch B replaces both the total and the output with the EB-2025-0297 settlement path: total regulated payments of C$5.36 billion on 52.2 TWh in 2027 rising to C$8.08 billion on 64.9 TWh in 2031, less market revenue at the same scenario-shifted leg; 2032 to 2035 hold the 2031 settled levels, an extension we label as ours, not the filing's.

GA-OPA has three parts. The schedule-priced subset carries its 2024 per-fuel contract totals (wind C$135.00/MWh, HCI hydro C$69.00/MWh, solar C$443.00/MWh on the Northland aggregate) against the scenario leg, with dated expiries from the roll-off schedule removing top-ups after their end years and expired wind, solar and waterpower output replaced at the LRP I reference prices (C$85.90, C$156.70 and C$175.90/MWh). Bruce enters at the bounded pair, C$65.73/MWh and about C$77/MWh, on flat 2024 output of 45.98 TWh. The residual pool, calibrated at C$3.20 billion (2024 GA-OPA less Bruce at the low bound less the subset), is held fixed in dollars and decayed by the cumulative dated-capacity expiry share of the GA-OPA book. GA-OEFC-NUG declines 10 percent per year from its 2025 actual of C$0.023 billion. The post-2021 tax-base perimeter is held constant throughout: the forecast is of GA as published on the current basis.

Market-leg scenarios are flat real paths for the covered-fleet generation-weighted leg, anchored on the derived history of 2015 to 2024 (which runs from C$13.30/MWh in 2020 to C$46.76/MWh in 2022): Low C$14.85/MWh (the mean of the trough years 2016, 2017 and 2020), Central C$31.39/MWh (the mean of 2023 and 2024), High C$46.76/MWh (2022). Facility legs shift by the scenario's difference from the 2024 fleet leg, so each facility keeps its 2024 position relative to the fleet, including the wind discount and the gas premium measured in Section 7.

The Class B rate is computed by the held identities. Class A takes a modelled share of Total GA and of consumption; Class B pays the rest over the rest. Two candidate models per share, a linear trend and the last-five-years level, were selected by holdout root mean squared error (train 2015 to 2020, test 2021 to 2025); the level won both (amount share: 1.01 points against 7.25 for the trend; consumption share: 4.01 points against 8.09), giving a Class A amount share of 16.86 percent and a consumption share of 29.96 percent, held flat. The consumption-share model is the noisy one: a four-point error on it moves the Class B rate by roughly 6 percent, and we report that sensitivity rather than hide it. The denominator is the APO Reference demand path. The APO's 2026 demand of 152.46 TWh sits 2.7 percent above the 2025 actual GA consumption base of 148.46 TWh; we use the APO path as published and note the wedge.

5.5 The backtest protocol

The forecast machinery is graded by running it as of end-2019, calibrated on 2019, forecasting 2020 to 2024 with the same rules (the class model refitted on 2015 to 2019 by the same selection protocol, which there selects the trend). Each year's actual fleet leg is fed in, so price error is excluded by construction and the residual error is machinery plus policy. No 2019-vintage APO path is held, so the backtest denominator is actual consumption. We report the error decomposition in Section 9.3 rather than a single score, because the decomposition is the finding.

6 Results I: the anatomy of the actual charge

6.1 Size and shape, 2005 to 2026

Figure 1 and Table 2 present the assembled series (Appendix A, Table 8, gives all years). GA began as a small and occasionally negative item: 2005 was a credit year at −C$1.15 billion, and 18 months of the 261 in the series are negative. It grew with the contract portfolio to C$9.96 billion in 2015, C$12.33 billion in 2016 and a peak of C$13.96 billion in 2020, when low market prices (the covered-fleet leg averaged C$13.30/MWh in 2020, the lowest of the derived history) maximised every top-up in the book. It then fell in two steps that were decisions, not market moves: to C$8.55 billion in 2021 on the tax-base shift, and within that new perimeter it has since run between C$5.27 billion (2025)

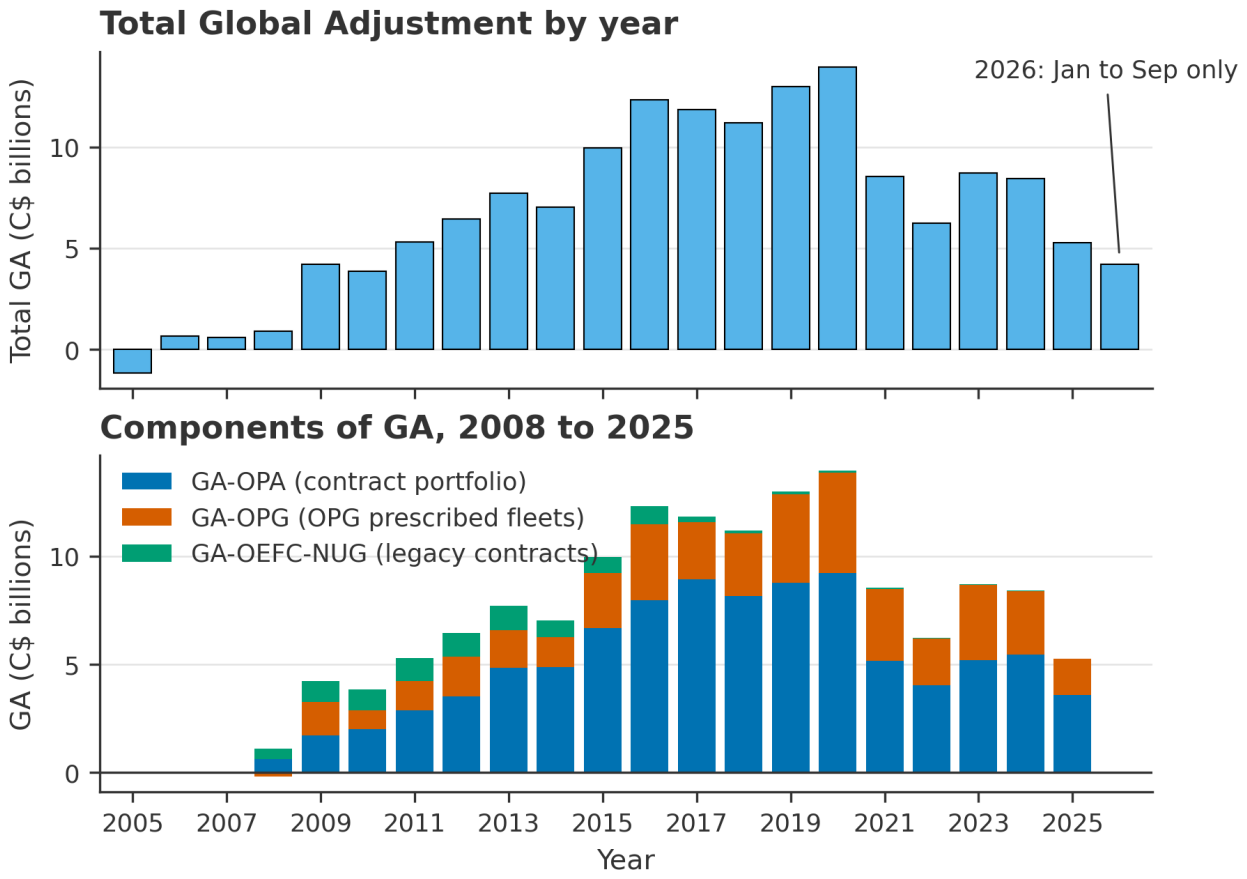


Figure 1: Total Global Adjustment and its three components, 2005 to 2026
Source: GA actual series, assembled from IESO GA publications (Observed, A). Note: Components are published from 2008; the components panel ends in 2025 because the 2026 component split is published only in the components workbook's next annual edition. GA-OPG for 2008 was negative (a credit of C$0.19 billion) and is drawn below the axis in the components panel. The 2026 total covers January to September.

and C$8.70 billion (2023), moving inversely with the market leg exactly as the contract arithmetic requires.

The ten-year comparison states the charge's scale in the units customers experience. Summing the monthly series from October 2016 to September 2026 gives Total GA of C$94.48 billion (Reconstructed, A). The same window's wholesale commodity value, from the estate's compilation of IESO settlement publications (which matches the assembled series' annual GA totals to C$0.1 million in every overlapping year), is C$47.11 billion. The ratio is 2.01: for every dollar the market paid for energy, the contract system collected two dollars on top. The ratio is perimeter-dependent, and we state its window for that reason: the window spans the 2021 tax-base shift, so it averages the charge on its old, wider basis with the charge on its current one.

6.2 Composition

GA-OPA is the centre of mass in every year: C$69.57 billion of the C$104.20 billion collected over 2015 to 2024, against GA-OPG's C$32.29 billion and GA-OEFC-NUG's C$2.34 billion. GA-OPG is the most stable pool in dollars and the most exposed to a single decision, since one rate case sets it

Table 2: Global Adjustment by component and the Class B rate, 2015 to 2025

Year	Total GA (C$ B)	GA-OPA (C$ B)	GA-OPG (C$ B)	GA-OEFC-NUG (C$ B)	Class B rate (C$/MWh)	Class A share of GA (%)
2015	9.96	6.67	2.57	0.72	77.77	11.0
2016	12.33	7.97	3.52	0.84	96.50	11.8
2017	11.85	8.93	2.65	0.27	98.20	14.1
2018	11.20	8.17	2.89	0.14	90.87	18.0
2019	12.98	8.76	4.11	0.12	107.66	18.4
2020	13.96	9.22	4.64	0.10	118.19	17.2
2021	8.55	5.15	3.34	0.06	72.43	16.8
2022	6.23	4.05	2.15	0.03	51.64	17.2
2023	8.70	5.20	3.48	0.03	73.60	17.2
2024	8.43	5.45	2.95	0.03	70.61	16.7
2025	5.27	3.59	1.66	0.02	42.59	16.4

Notes: The three components sum to Total GA in all 132 months (verified to the cent in the build). The Class B rate is the Class B amount divided by Class B consumption, which matches the IESO published actual rate except in the policy windows named in Section 6. The Class A share is Class A dollars over Total GA. GA before 2021 and after 2021 is not on one basis: from January 2021 the published GA-OPA excludes about 85 percent of non-hydro renewable contract costs, which moved to the tax base. *Source:* IESO GA publications as assembled in the GA actual series (Observed, A).

for five years (Section 9). GA-OEFC-NUG shows the terminal state of a contract pool left to expire: C$0.72 billion in 2015, C$0.03 billion in 2024, and a forecast assumption of continued decay that no scenario in Section 9 needs to argue about.

6.3 The Class B rate and the estimate record

The Class B rate (Table 2) peaked at C$118.19/MWh in 2020, fell to C$51.64/MWh in 2022 as market prices spiked and top-ups compressed, and reached C$42.59/MWh in 2025, the lowest of the workbook era. The rate's level is a ratio, and both sides move: the numerator fell on the tax-base shift while Class B consumption recovered from 97.8 TWh in 2020 to 103.5 TWh in 2025.

Two features of the record matter for the forecast. First, GA is estimated before it is settled, on a published ladder of first estimate, second estimate and actual. The held record of that ladder (the components workbook's second-estimate and actual rows, 2018 to 2025) shows annual sums of monthly second estimates landing within −0.1 percent to +4.5 percent of the annual actual, while at monthly grain the middle 80 percent of second-estimate errors (second estimate less actual, over actual) ran from −6.9 percent to +13.8 percent, and months whose actuals are near zero make percentage errors unbounded (December 2025's actual was a credit of C$34.4 million). That is the error on estimating a known system one month out; it is a floor under any forecast's first-year uncertainty, not a band around it. Second, the renewed market's first year is wider than the old record: January and February 2026 settled as large credits after positive estimates, and 2026 monthly actual Class B rates run from −$40.62/MWh to C$99.51/MWh. A forecast that reports a point estimate for 2027 without a range has not absorbed its own data.

Table 3: Generation, market leg, and contract top-up by fuel, 2024

Fuel	Generation (TWh)	Market leg (C$/MWh)	Totals coverage (%)	Contract total on subset (C$/MWh)	Top-up on subset (C$/MWh)
Nuclear	79.44	32.55	0.0	—	—
Hydro	36.24	33.48	4.4	69.00	33.07
Gas	25.31	37.85	0.0	—	—
Wind	13.39	28.76	44.9	135.00	106.28
Solar	0.69	33.57	9.0	443.00	408.00
Bioenergy	0.29	37.48	0.0	—	—

Notes: Generation and market legs cover the facilities in the derived table (99.03 percent of 2024 system generation of 156.88 TWh). Contract totals exist only for the subset of facilities whose prices are public; the coverage column states that subset's share of the fuel's covered generation, and the total and top-up columns describe the subset only. The solar subset is one reporting aggregate at early feed-in tariff schedule prices; its top-up is not a solar fleet average. Nuclear and gas have no facility totals in the public record. *Source:* Fuel price panel, calendar 2024 (Reconstructed, A).

7 Results II: facility-level evidence

7.1 Market legs by fuel

Table 3 gives the 2024 panel, and Figure 2 draws the monthly pattern over 2018 to 2023, the longest window in which every series is continuous on a single price concept (HOEP). Three regularities stand out.

Wind earns less than the fleet. In every year from 2010 to 2024 the wind fleet's output-weighted leg sat below the covered-fleet leg: by 9.5 percent in 2010, 25.6 percent in 2020, and 13.7 percent in 2024 (Reconstructed, A). At the FIT subset total of C$135.00/MWh the measured wind top-up is C$106.28/MWh (the subset covers 44.9 percent of the wind fleet's 13.39 TWh), against a fleet-average price gap that a naive calculation, the same total less the fleet leg of C$33.32/MWh, would put C$4.60/MWh lower. The discount is the capture effect of Hirth (2013) measured on settlement-grade data, and it means GA's wind line is structurally larger than wind's share of energy suggests.

Gas earns more than the fleet, and holds no public total. The gas fleet's 2024 leg of C$37.85/MWh is the highest of the large fuels, and the premium is persistent at facility level: Goreway Station's 2020 leg of C$19.82/MWh stood 49 percent above the covered-fleet leg of C$13.30/MWh in the cheapest year of the record. Gas contracts in Ontario are predominantly net-revenue-requirement and cost-of-service forms (CES, ACES, CHP) under which no per-megawatt-hour energy price exists to publish; the register therefore carries no gas facility total, and the fuel panel's gas top-up is not zero, it is unobservable at facility level. Section 7.3 returns to what that does to the reconciliation.

Solar's measured top-up is a coverage artefact and we treat it as one. The only solar total in the public record is one reporting aggregate (13 facilities, 130 MW) at the early FIT schedule price of C$443.00/MWh, covering 9.0 percent of covered solar generation in 2024. Its C$408.00/MWh top-up describes that aggregate's contracts. It must not be read as a solar fleet average, and the forecast's solar line is correspondingly small in dollars (the subset generated 0.06 TWh in 2024 on the totals measure).

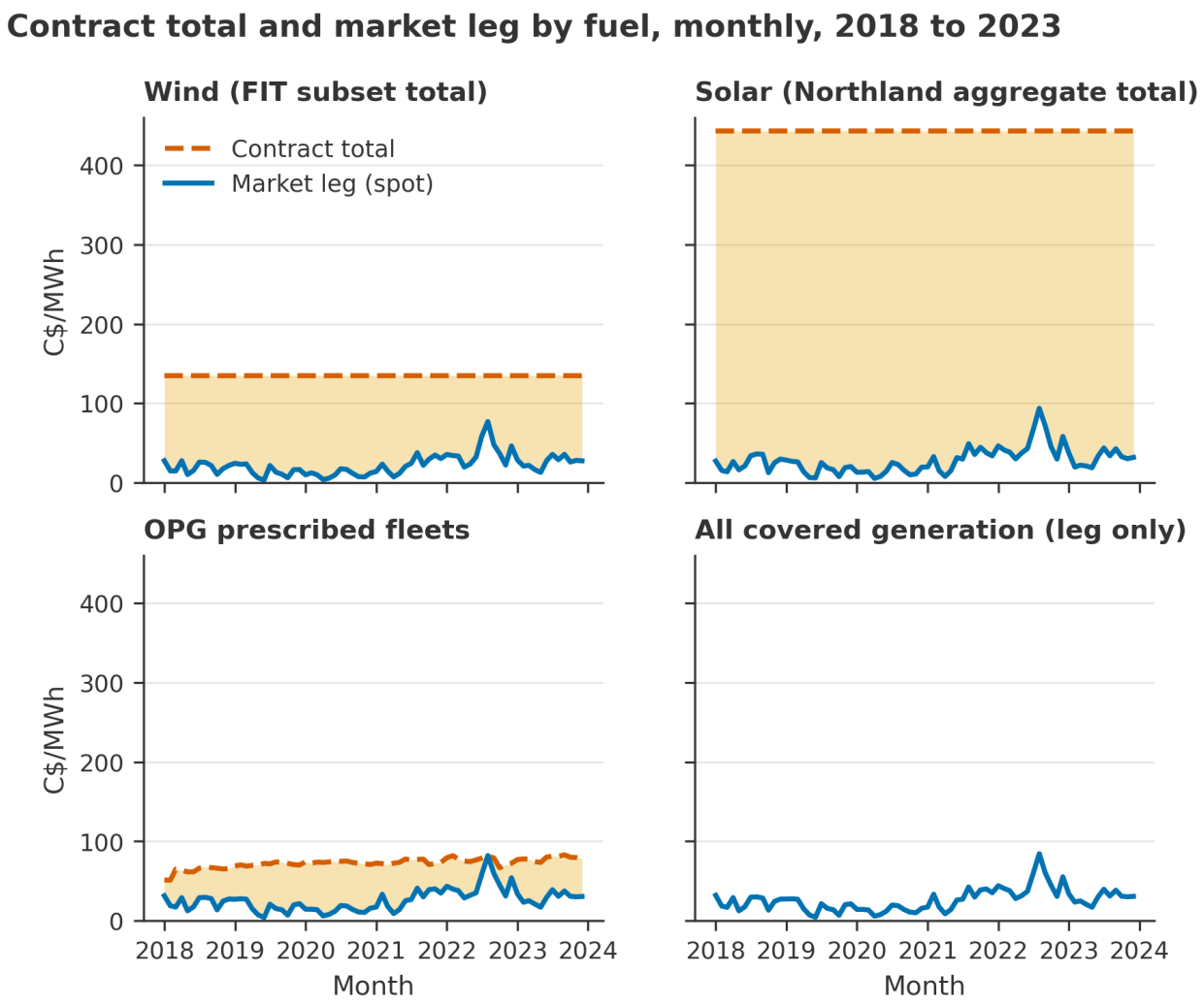


Figure 2: Contract total and market leg by fuel, monthly, 2018 to 2023

Source: Fuel price panel and derived market rates (Reconstructed, A). Note: All four panels share one vertical scale so the solar total reads against the same axis as the others. The wind total is the FIT schedule price of C$135.00/MWh, covering 45.7 percent of covered wind generation over this window; the solar total is one reporting aggregate at an early FIT price of C$443.00/MWh, covering 9.6 percent of covered solar generation over the window (1.1 to 18.0 percent by month), and is not a solar fleet average. The OPG total is implied from GA-OPG for the two prescribed fleets combined. The shaded band between total and leg is the top-up GA recovers. The window is HOEP throughout; no series crosses the May 2025 to July 2026 price gap.

Table 4: Facility-level reconstruction of GA against actual GA, 2015 to 2024 (C$ billions)

Year	Modelled OPG	Modelled subset	Modelled total	Actual Total GA	Explained (%)	Subset as % of GA-OPA
2015	2.57	0.49	3.06	9.96	30.7	7.3
2016	3.52	0.61	4.12	12.33	33.4	7.6
2017	2.65	0.69	3.33	11.85	28.1	7.7
2018	2.89	0.66	3.56	11.20	31.8	8.1
2019	4.11	0.73	4.84	12.98	37.3	8.4
2020	4.64	0.86	5.50	13.96	39.4	9.3
2021	3.34	0.70	4.04	8.55	47.3	13.7
2022	2.15	0.68	2.83	6.23	45.4	16.8
2023	3.48	0.70	4.18	8.70	48.0	13.5
2024	2.95	0.72	3.67	8.43	43.5	13.1
2015–2024	32.29	6.84	39.13	104.20	37.6	9.8

Notes: The modelled OPG column is GA-OPG by construction: the combined prescribed-fleet row's top-up is defined from the published GA-OPG component, so its agreement is a build check on the output and price machinery, not independent validation. The subset is the schedule-priced facilities (HCI hydro, feed-in tariff wind with output feeds, one solar aggregate). The unexplained balance sits inside GA-OPA and GA-OEFC-NUG; Section 7.3 names its largest blocks. From 2021 the published GA-OPA excludes about 85 percent of non-hydro renewable contract costs while the modelled subset uses full schedule prices, so post-2021 explained shares mix bases. *Source:* Modelled columns: derived market rates and fuel price panel (Reconstructed, A). Actual column: GA actual series (Observed, A).

7.2 What the top-ups cost at 2024 volumes: two mix shifts

The panel supports accounting counterfactuals at fixed prices. On a 2024 system basis (156.88 TWh; modelled GA generation component C$6.45 billion; blended all-in fleet cost C$74.43/MWh), shifting 10 percent of total generation (15.69 TWh) from gas to wind at the measured top-ups raises modelled GA by C$1.67 billion per year and the blended all-in cost by C$9.72/MWh, to C$84.15/MWh. Shifting the same volume from nuclear to gas lowers modelled GA by C$0.68 billion per year and the blended cost by C$3.82/MWh, to C$70.61/MWh (Modelled, B). Both are arithmetic on the existing contract book, not build plans: new wind would price near the LRP I average of C$85.90/MWh, not the legacy FIT C$135.00/MWh that drives the first result, and the nuclear top-up used (C$43.50/MWh) is the OPG combined fleets' GA-implied figure, a nuclear-plus-hydro blend. What the shifts establish is the sign and rough scale of the mix term in any GA forecast: at legacy prices, mix is a billion-dollar lever, and its direction depends on which legacy contracts the new energy displaces.

7.3 The reconciliation: 43.5 percent explained, the rest named

Table 4 sets the facility reconstruction against actual GA for 2015 to 2024. The explained share rises from 28 to 33 percent in 2015 to 2017 to 43.5 percent in 2024 (48.0 percent in 2023); monthly, the explained share has a median of 38.2 percent and a range of 13.4 to 61.5 percent over 120 months. Over the decade the reconstruction explains C$39.13 billion of C$104.20 billion.

The unexplained C$4.76 billion in 2024 (56.5 percent of that year's GA) decomposes by pool, not by mystery. GA-OEFC-NUG contributes C$0.03 billion, unmodelled because the OEFC publishes dollars without volumes or membership. The rest, C$4.73 billion, is GA-OPA beyond the schedule-priced

subset, which is 13.1 percent of GA-OPA in 2024. Inside that remainder the blocks are known even where they cannot be priced facility by facility:

Bruce Power's top-up is the largest single block. Bruce produced 45.98 TWh in 2024 at a derived leg of C$32.53/MWh (RECONSTRUCTED, A). At the documented 2016 base price of C$65.73/MWh its top-up is C$1.53 billion; at the stated life average of about C$77/MWh it is C$2.04 billion. Even the low bound is 28 percent of 2024 GA-OPA. The true figure also carries CPI indexation and refurbishment step adders, which no held source prices by year, so the paper carries Bruce as a bounded input everywhere, including the forecast.

The pooled programs are the second block: RES, LRP, CES, ACES and GEIA contracts whose prices are disclosed at program level or not at all (AGGREGATE-ONLY in the register), plus the embedded FIT fleet's top-ups, whose facilities appear in no output series. These are real GA dollars attached to exactly the register rows the coverage accounting of Section 4 cannot price.

One basis warning applies to every post-2021 row of Table 4: published GA-OPA after the tax-base shift excludes about 85 percent of non-hydro renewable contract costs, while the modelled subset is computed at full schedule prices. The post-2021 explained shares therefore mix a gross modelled numerator with a net published denominator, and are, if anything, flattered.

The identity itself is not in doubt. Where totals are public, the machinery reproduces published dollars: the OPG calibration of Section 5.2, the component sums of Section 6, and the subset's behaviour across 120 months all point the same way. The binding constraint on explaining GA is the disclosure structure of the pooled programs, and that is a finding about the public record, not about the charge.

8 Results III: the roll-off schedule

The register's own term-end dates give the expiry path of the contract book. An end year is determinable for 3,553 of 3,582 register rows (99.2 percent) and for 28,396 MW of the 32,428.8 MW stated (87.6 percent). The 29 undated rows, 4,032.8 MW, are all under-development capacity contracts (LT 1: 2,164.2 MW; LT 2: 1,755.1 MW; ELT 1: 109.5 MW; one 4.0 MW PPA) whose register entries carry no term dates in any field; no end year was assumed for them. Figure 3 and Table 5 show the shape.

By capacity, two years dominate the window. In 2035, 5,431.6 MW ends: the negotiated gas fleet (CES 1,685.5 MW and ACES 1,552.5 MW), GEIA contracts of 650.0 MW, the York Energy Centre (NYRP, 431.0 MW) and FIT 1 contracts of 653.2 MW. In 2029, 4,805.8 MW ends: Lennox (LESA, 2,000.0 MW), the HCI waterpower fleet (974.2 MW), RES II wind (702.5 MW) and Portlands (ACES, 600.0 MW). Bruce Units 1 and 2 (1,646.0 MW) end in 2043, and the refurbished Units 3 to 8 sit in the 2046+ bucket.

By dollars, the cliff is later and smaller, because the facilities whose top-ups can be computed are mainly the FIT fleet. The largest quantified year is 2034 at C$152.58 million per year (568 rows, the FIT 1 wave), then 2035 at C$129.69 million, 2033 at C$122.18 million and 2036 at C$109.86 million. Across all end years the quantified expiring top-up is C$716.2 million per year at 2024 volumes. That figure is a floor, for three named reasons: Bruce's 6,549 MW rolls off with no computable dollars (Section 7.3); the AGGREGATE-ONLY fleets (13,799.8 MW dated) have no per-facility prices in the public record; and about 3,370 tariff facilities beyond the 36 with output feeds hold schedule prices but no facility output series, so their top-ups cannot be computed per facility. What the floor establishes is the minimum mechanical decay built into GA-OPA's priced core through the 2030s: on

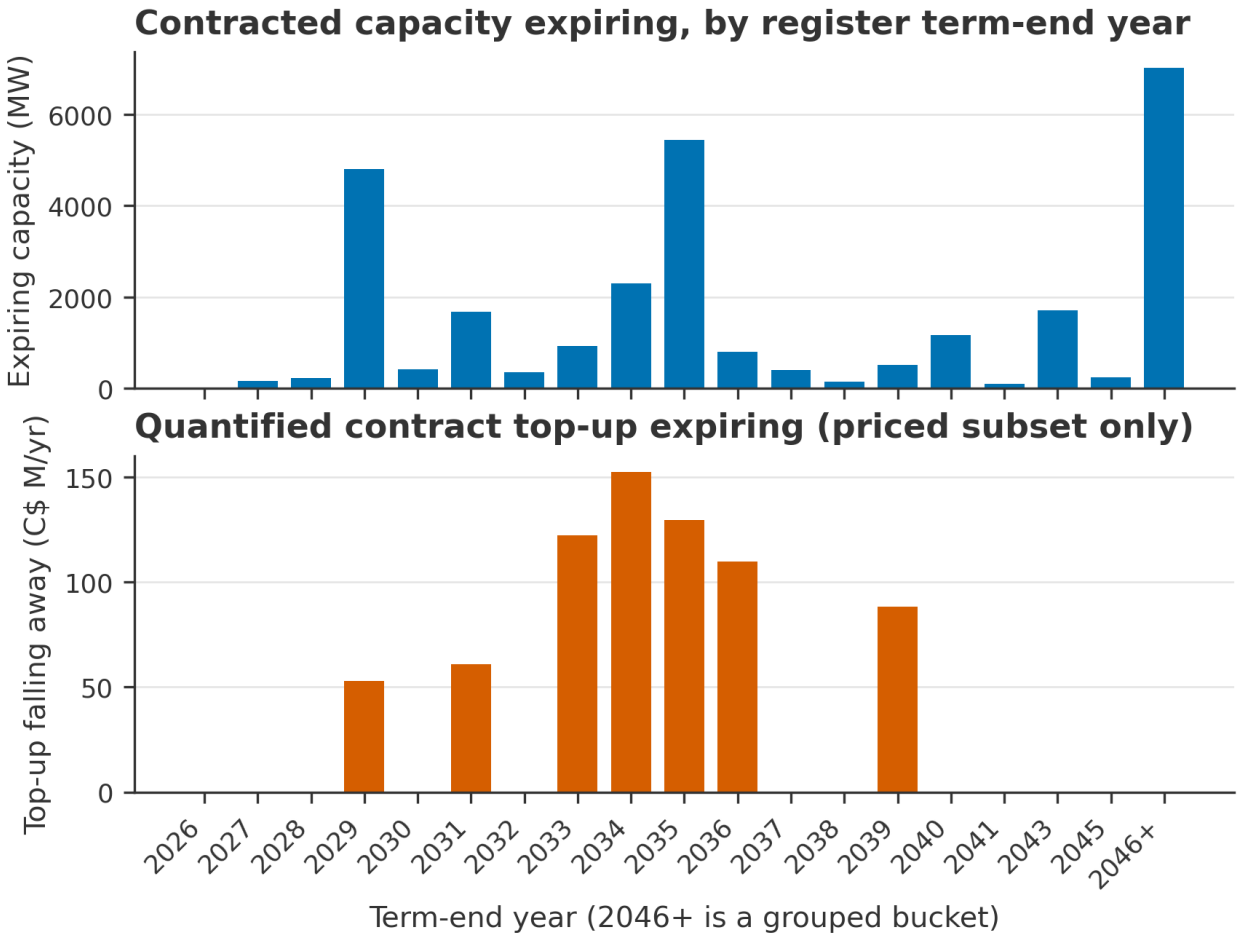


Figure 3: Contract roll-off by register term-end year
Source: Contract roll-off schedule from the register (Observed, A for capacity; Reconstructed, A for top-up dollars). Note: The 2046+ bucket groups all later end years, including Bruce Units 3 to 8 (4,903.0 MW, ending 2058 to 2063) and HESA waterpower on 50-year terms (1,063.4 MW). Years with no bar in the lower panel have no facility with a derived contract total ending that year; that is a statement about price coverage, not about GA. The OPG prescribed fleets are excluded from both panels: their 2026 entry is a rate reset, not an expiry.

the order of C$0.72 billion per year of top-up falls away between 2029 and 2039 if nothing replaces it.

Replacement is the offsetting term, and the held record prices it for one procurement. The IESO's LRP I results (March 2016), the only procurement-results document in the vault, give weighted average contract prices of C$85.90/MWh for wind (range C$64.50 to C$105.50), C$156.70/MWh for solar and C$175.90/MWh for waterpower. Against the expiring FIT wind total of C$135.00/MWh, like-for-like replacement at the LRP I wind price cuts the replaced volume's top-up by about half at the 2024 wind leg (from C$106.28/MWh to C$57.14/MWh); against the Northland solar total of C$443.00/MWh the cut is far deeper. Two cautions attach. LRP I prices are 2016 prices for 20-year terms, and using them for 2030s replacement without a price-path assumption understates replacement cost. And the newer capacity procurements (LT 1, LT 2, ELT 1) pay for capacity in C$/MW-month, not energy: no LRP-form energy replacement price exists for them, and no LT 1 or ELT 1 results document is held. The forecast in Section 9 therefore replaces only the quantified renewable subset at LRP I prices and treats the rest of the expiring book through the residual pool's decay, which is the honest limit of the held record.

Table 5: The roll-off: largest term-end years in the contract register

Term-end year	Register rows	Expiring capacity (MW)	Quantified top-up falling away (C$ M/yr)
2029	54	4,805.8	52.75
2031	220	1,680.6	60.91
2033	381	918.9	122.18
2034	568	2,303.8	152.58
2035	691	5,431.6	129.69
2036	531	805.0	109.86
2039	30	510.5	88.27
2043	12	1,703.8	not quantified
2046+	55	7,012.5	—
All end years	3,553	28,396	716.2

Notes: Top-up dollars are at 2024 volumes and 2024 top-ups and cover only facilities with a derived contract total (36 tariff facilities plus the OPG combined row): the all-years figure of C$716.2M per year is a floor for the priced fleet's roll-off, not its total. Bruce (2043 and the 2046+ bucket) carries capacity and no dollars because no year-specific Bruce total is held. The OPG prescribed fleets do not appear: their current rate period ends 31 December 2026 as a rate reset, not an expiry (Section 8). *Source:* Contract roll-off schedule built from the register's own term-end dates (Observed, A for dates and capacity; Reconstructed, A for top-up dollars).

The OPG fleets stand outside the roll-off as a rate reset. Their current OEB rate period ends 31 December 2026 with, at 2024 volumes and the 2024 GA-implied top-up of C$43.50/MWh, about C$2.95 billion per year of GA-OPG top-up flowing under it. None of that falls away in 2027; it reprices, under EB-2025-0297, which is why the forecast carries the case as a branch rather than an expiry.

9 Results IV: the forecast, 2027 to 2035

9.1 The grid

Table 6 and Figure 4 present the forecast as what it is: a grid of 108 cells (9 years, 3 market-leg scenarios, 2 OPG branches, 2 Bruce price bases), not a point prediction. Total GA in 2027 spans C$6.79 billion to C$11.58 billion; in 2035, C$5.38 billion to C$12.70 billion. The Class B rate spans C$51.54/MWh to C$87.94/MWh in 2027 and C$32.68/MWh to C$77.20/MWh in 2035 (Modelled, B).

The drivers rank clearly. The market-leg scenario is the largest: at branch A with Bruce at the low bound, the 2027 spread between the Low and High legs is C$3.88 billion, because every top-up in the book moves against the leg one for one. The OPG branch is second and grows: branch B less branch A on the central path is C$0.66 billion in 2027 and C$2.98 billion from 2031 on. The Bruce price basis is third at C$0.52 billion (the C$11.27/MWh between the documented bounds, on 45.98 TWh).

9.2 The two branches tell opposite stories

On the central path under branch A, Total GA declines from C$8.65 billion in 2027 to C$7.24 billion in 2035, and it declines in all six branch A cells: the calibrated pools are flat to decaying (the roll-off

Table 6: Forecast Global Adjustment, 2027 to 2035: ranges and central paths

Panel A. Ranges across the scenario grid (3 market-leg scenarios × 2 Bruce price bases)

	Branch A: current OPG economics		Branch B: EB-2025-0297 settlement	
Year	Total GA (C$ B)	Class B (C$/MWh)	Total GA (C$ B)	Class B (C$/MWh)
2027	6.79–11.18	51.54–84.92	7.68–11.58	58.36–87.94
2030	6.55–10.95	45.25–75.62	8.54–12.72	58.98–87.85
2035	5.38–9.77	32.68–59.39	8.40–12.70	51.07–77.20

Panel B. Central paths (Central market leg, Bruce at C$65.73/MWh)

	Branch A		Branch B	
Year	Total GA (C$ B)	Class B (C$/MWh)	Total GA (C$ B)	Class B (C$/MWh)
2027	8.65	65.72	9.31	70.71
2028	8.62	63.31	9.98	73.26
2029	8.58	61.54	10.16	72.86
2030	8.42	58.15	10.30	71.15
2031	8.35	55.86	11.33	75.78
2032	8.02	51.79	11.00	71.03
2033	7.96	50.54	10.94	69.46
2034	7.73	48.41	10.71	67.07
2035	7.24	44.03	10.22	62.14

Notes: Branch A holds OPG's 2024 implied regulated economics flat. Branch B substitutes the EB-2025-0297 settlement path for GA-OPG (settled payment amounts and production forecasts, 2027 to 2031; 2031 levels held to 2035, an extension the filing does not make). Market-leg scenarios are flat real paths: Low C$14.85/MWh, Central C$31.39/MWh, High C$46.76/MWh (Section 5.4). The settlement was unapproved at the build date; if the Ontario Energy Board's decision moves the amounts, branch B must be re-based (Section 11). *Source:* GA forecast grid (Modelled, B).

of Section 8 removes the priced subset's top-ups through the 2030s and the residual pool decays with dated expiries), while APO Reference demand grows 25.0 percent, from 156.31 TWh in 2027 to 195.33 TWh in 2035. Nothing in branch A adds a new contract: the model's only additions are LRP-priced replacements of the quantified renewable subset.

Under branch B the total rises on the central path from C$9.31 billion in 2027 to a peak of C$11.33 billion in 2031, and it rises from 2027 to 2035 in every branch B cell. The settled OPG revenue path is the cause: total regulated payments grow from C$5.36 billion on a refurbishment-trough production forecast of 52.2 TWh in 2027 to C$8.08 billion on 64.9 TWh in 2031, as the Pickering refurbishment return and the Darlington New Nuclear Project's rate base enter the payment amounts. Branch B's GA-OPG on the central path is C$3.74 billion in 2027 and C$6.06 billion in 2031, against branch A's flat C$3.08 billion. The near-term difference between the branches is modest because the settlement pairs much higher payment amounts (a weighted average of C$102.79/MWh against C$77.75/MWh approved for 2026) with a low 2027 production forecast; the difference widens as production recovers into the higher amounts.

The Class B rate falls in every cell of the grid, including branch B's. Class B consumption grows with demand while the Class A shares are held at their 2021 to 2025 means, so demand growth outruns even the settlement's revenue growth: on the central path the rate falls from C$65.72/MWh to C$44.03/MWh under branch A and from C$70.71/MWh to C$62.14/MWh under branch B (peaking at C$75.78/MWh in 2031). A rising GA total and a falling GA rate are consistent, and the grid's

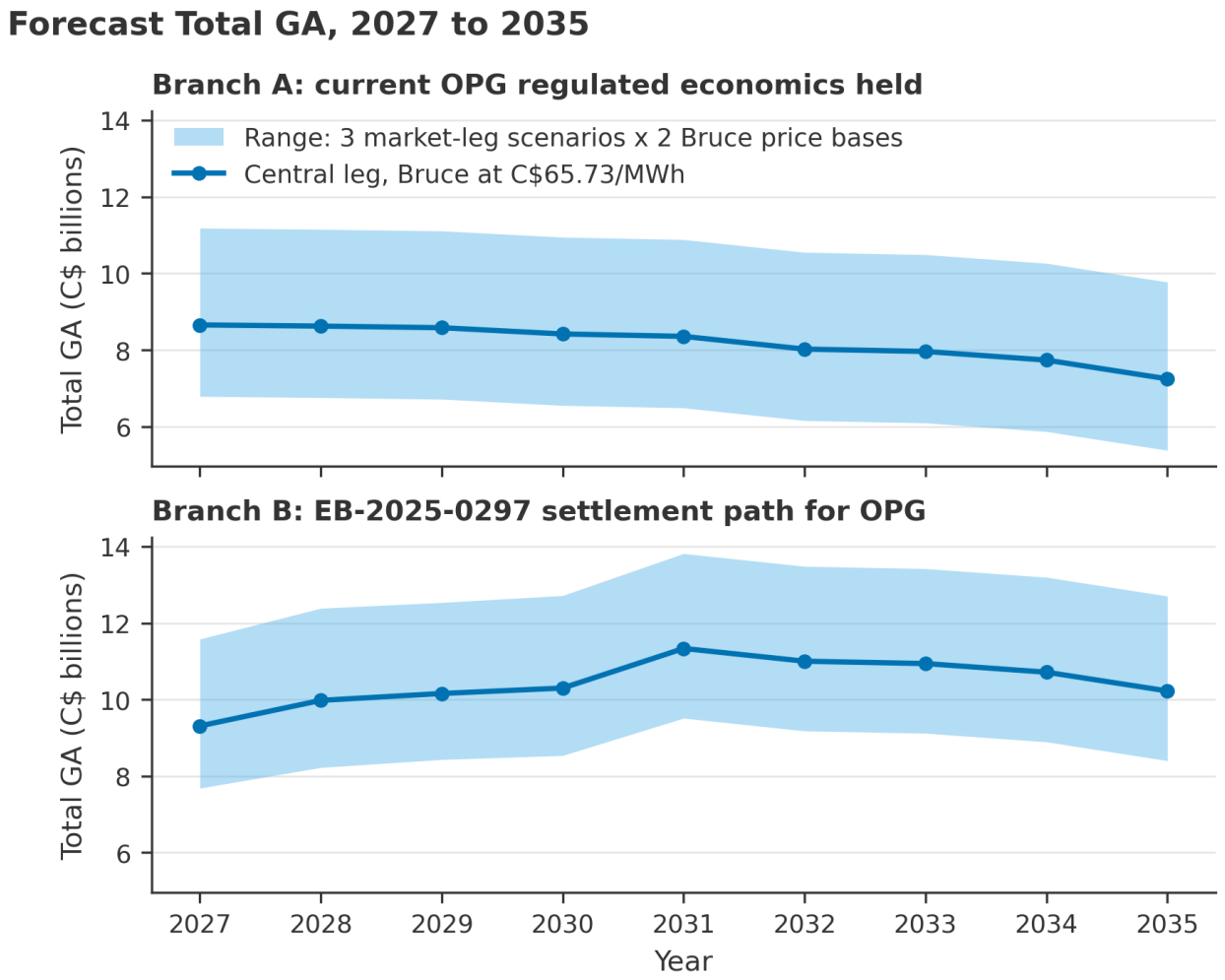


Figure 4: Forecast Total GA by OPG branch, 2027 to 2035
Source: GA forecast grid (Modelled, B). Note: Bands span the three market-leg scenarios and two Bruce price bases within each branch; the line is the central path (Central leg, Bruce at C$65.73/MWh). Branch B's 2032 to 2035 cells hold the settlement's 2031 revenue and production levels flat, an extension of this build, not a filing figure. The vertical scale is shared across panels and starts at C$5 billion; the axis shows the real range.

central message is that Ontario can have both for a decade if demand grows as the APO projects.

9.3 The backtest: the machinery is close, the perimeter is the risk

Table 7 and Figure 5 report the end-2019 backtest. Cumulatively the model overstates 2020 to 2024 GA by C$10.80 billion (+23.5 percent). The decomposition, not the total, is the result:

In 2020, the one year with no policy change to the GA perimeter, the machinery errs by −C$0.43 billion (−3.1 percent) on Total GA and by −C$1.50/MWh (−1.3 percent) on the Class B rate. Given true prices, the pool machinery is close.

The 2020 pandemic deferral explains none of the annual error: the Class B cap at C$115.00/MWh for April to June 2020 shifted charges within the year and into 2021's recovery rates, and at annual grain it nets out. Its signature is monthly, and this model is annual; we name it because a monthly reading of 2020 would otherwise misattribute it.

Table 7: Backtest: the forecast machinery as of end-2019, forecasting 2020 to 2024

Year	Model Total (C$ B)	Actual Total (C$ B)	Error (C$ B)	Error (%)	Model Class B (C$/MWh)	Actual Class B (C$/MWh)
2020	13.53	13.96	-0.43	-3.1	116.70	118.19
2021	11.63	8.55	+3.08	+36.0	100.86	72.43
2022	9.19	6.23	+2.96	+47.5	77.83	51.64
2023	11.41	8.70	+2.71	+31.1	97.54	73.60
2024	10.91	8.43	+2.48	+29.4	91.12	70.61
2020–2024	56.67	45.87	+10.80	+23.5	—	—

Notes: The backtest is fed each year's actual fleet market leg, so price-forecast error is excluded by construction; what remains is machinery error plus policy. The 2021 tax-base shift accounts for most of the post-2020 error (Section 9.3). No 2019-vintage APO demand path is held, so the backtest denominator is actual consumption. *Source:* Backtest build (Modelled, B) against the GA actual series (Observed, A).

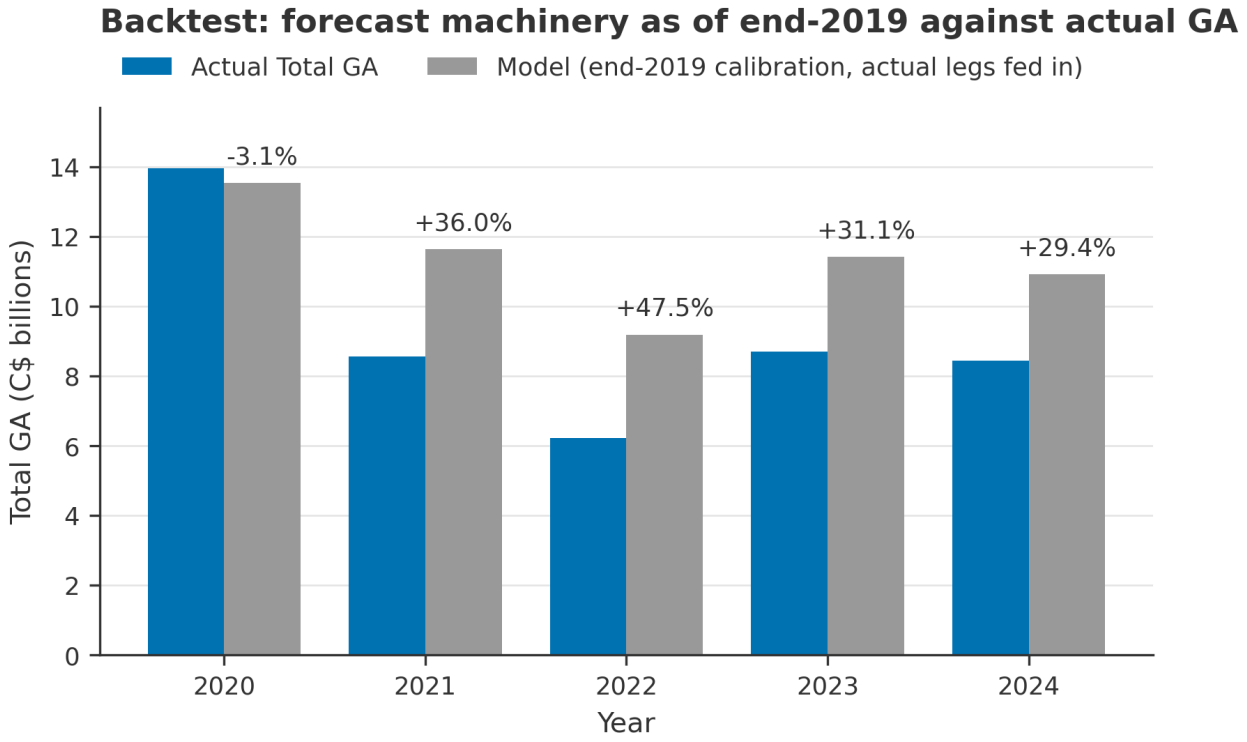


Figure 5: Backtest: model against actual Total GA, 2020 to 2024
Source: Backtest build (Modelled, B) and GA actual series (Observed, A). Note: Percentage labels are the model's error against the actual for that year. The backtest is fed actual market legs, so the errors shown exclude price-forecast error by construction.

The 2021 tax-base shift is the error. The model's error steps by +C$3.51 billion between 2020 and 2021, the first-year fingerprint of actual GA-OPA falling from C$9.22 billion to C$5.15 billion as about 85 percent of non-hydro renewable contract costs left the charge. The model, calibrated on the gross basis in the same way the forward build holds the current perimeter as an explicit assumption, keeps carrying the shifted costs, and that persistence accounts for most of the 2022 to 2024 errors of +C$2.96 billion, +C$2.71 billion and +C$2.48 billion. Inside the persistence, three named items

partly offset: Bruce's refurbishment-era output ran 3.5 to 3.8 TWh per year below the flat-2019 assumption in 2021 to 2023, overstating the modelled Bruce top-up by C$0.51 billion cumulatively; the NUG decline rule overshoots by C$0.18 billion cumulatively; and the GA-OPG machinery runs C$0.80 billion cumulatively under actuals, because OPG's implied totals stepped up with the 2022 to 2026 rate period in a way a flat calibration cannot see. The remainder is residual-pool drift: contract entries, refurbishment cost steps and re-pricings inside GA-OPA that a flat pool cannot represent.

Read forward, the backtest says the grid in Table 6 should be trusted in a specific way. Conditional on the perimeter and on a market-leg path, the pool machinery is accurate to a few percent. The dominant risk to every cell is a future perimeter decision, moving costs into or out of GA by policy, which no cell prices. The 2026 estimate record of Section 6.3 adds a second caution: even the first year carries the renewed market's wider true-up variance.

10 Discussion

Three implications follow from the measurement, each stated at the strength the evidence supports.

GA is a stock variable wearing a flow variable's clothes. Its annual total is read in public debate as a comment on the current government or the current market. The decomposition shows it is neither: it is the running cost of a contract book assembled between the 1990s and the 2010s, repriced by the market leg and altered only at the margin by current decisions, except when a decision changes the perimeter itself. The two largest moves in the 22-year series, the 2017 Fair Hydro reduction and the 2021 tax-base shift, were perimeter and financing decisions that changed what the charge contains and when it is paid, not what the contracts cost. Any assessment of a year's GA that does not first ask which basis the year is on will misread it; the series in Section 6 is built to make that question unavoidable.

The disclosure structure is the binding constraint on accountability, and it binds asymmetrically. Ontario publishes enough to price 99.03 percent of generation at facility level and to verify the charge's arithmetic exactly at pool level. It does not publish the per-facility prices of the pooled programs, which is why 56.5 percent of 2024 GA cannot be attributed below pool level from public data, and why the largest single contract in the system, Bruce's, enters every serious calculation as a bound rather than a number. The gap is not a data-processing problem awaiting a better model. It is the disclosure design of the programs themselves (pooled, program-level, or in Bruce's case settled inside a pool and never itemised), and it has a measurable price: the difference between the Bruce bounds alone, C$0.52 billion per year at 2024 output, is larger than the quantified roll-off of the entire priced FIT fleet in any single year. Debates about GA's level proceed, at present, with the largest single component known only within a band of that width.

The forecast's practical content is the branch, not the range. Market-price scenarios move Total GA by up to C$3.88 billion in 2027, but they move it symmetrically and temporarily: a high-price year compresses top-ups that a low-price year restores. The EB-2025-0297 branch moves GA persistently and in one direction, because regulated payment amounts, once set, apply for five years regardless of the market. On the central path the settlement reverses the sign of GA's drift for a decade: a charge that was decaying toward C$7.24 billion by 2035 instead peaks at C$11.33 billion in 2031 and ends the window at C$10.22 billion. The Class B rate falls under both branches only because demand is projected to grow 25.0 percent; if APO demand growth disappoints, the rate arithmetic reverses faster than the totals arithmetic, since the Class A allocation concentrates the residual on Class B. Readers who take one number from Section 9 should take the pairing, not either element: totals are a contract-book story, rates are a contract-book story divided by a demand story.

11 Limitations and robustness

The limitations are stated at the same prominence as the results, because several of them are the results' boundaries.

The unexplained share is large and structural. The facility reconstruction explains 43.5 percent of 2024 GA. The balance is attributable to named pools (Section 7.3) but not to facilities, and no extension of the present method closes it: the pooled programs do not disclose facility prices, the embedded FIT fleet appears in no output series, and Bruce's year-specific total is not published. Conclusions that depend on pool totals (the anatomy of Section 6, the forecast of Section 9) rest on published aggregates and are not weakened by this limit. Conclusions that would depend on facility attribution inside GA-OPA cannot be drawn from the public record, and we do not draw them.

Bruce is a bound, not an estimate. Every Bruce dollar in the paper is computed at one of two documented prices, C$65.73/MWh (the 2016 base) and about C$77/MWh (the stated life average). The true path carries CPI indexation each April and refurbishment step adders, so the true top-up in any year can sit inside or, in principle, outside the printed band. The forecast carries both bases through every cell so the sensitivity is visible rather than assumed away: C$0.52 billion per year at 2024 output.

The price gap fixes the forecast's anchor year. No market prices are held for May 2025 to July 2026, so the most recent full-year leg anchor is 2024 and the scenario paths are anchored on the HOEP-era distribution. The renewed market's first months print higher legs than the HOEP-era norm (the covered fleet's monthly legs for August to November 2026 run from about C$48/MWh to C$61/MWh on the nodal concept). If that level persists, the Low and Central paths are too low. The legs are also different price concepts across the renewal (a uniform provincial price against delivery-point prices), and we do not splice them into one series anywhere in the paper.

The forecast holds volumes and mix constant. Prescribed-fleet output (branch A), Bruce output, subset volumes and the generation mix are held at 2024, because the APO workbook carries no generation-by-fuel outlook for grid-connected supply. The fuel build's counterfactuals bound the mix error at 2024 volumes (Section 7.2): a 10-point mix shift moves modelled GA by C$0.68 billion to C$1.67 billion per year. Branch B relaxes the volume assumption for OPG only, using the filing's production forecasts. New-contract additions beyond LRP-priced replacement of the quantified subset are excluded; the LT and ELT capacity fleets carry no energy price in the form the model uses, and if their costs enter GA in a form the actuals do not yet show, the model misses them.

The residual pool is the model's weakest part. It is held fixed in dollars across leg scenarios, which suppresses its own price sensitivity and compresses the grid's leg-driven spread (GA-OPA is biased high in High-leg cells and low in Low-leg cells). Its decay is proportional to dated megawatts, not to dollars or energy, and re-contracting of decayed pool contracts, the historical norm for gas and waterpower, is not modelled, which biases GA-OPA down in the 2030s if re-contracting continues. The backtest's residual-pool drift (Section 9.3) is the measured size of this weakness over five years.

Branch B is a settlement, not a decision. The OEB may move the settled amounts through the return-on-equity update the parties agreed to accept, and the excluded C$170.2 million CRVA balance would add a further nuclear rider, from July 2027 at the earliest, if allowed. Branch B's 2032 to 2035 cells extend the settlement's 2031 levels; DNNP output in reality ramps beyond 2031 and a successor application will reset the path. Branch B should be re-based when the Decision and Order issues, and the paper's branch B numbers carry that expiry.

Register totals are base schedule rates. Escalation and CPI indexation recorded as register text are not arithmetically applied to any facility total, so expiring top-ups at 2030s prices will differ from

the 2024-basis figures of Section 8 by the indexation not applied. The roll-off dollar floor is therefore a floor in two senses.

Class A modelling noise passes into the Class B rate. The Class A consumption share is held at its 2021 to 2025 mean (29.96 percent); its holdout root mean squared error is 4.01 percentage points, and a four-point error moves the Class B rate by roughly 6 percent in every forecast year, on top of the numerator's range.

Falsifiability

The paper's main conclusions would be falsified, in whole or in part, by any of the following. If the IESO publishes a program-level split of GA-OPA showing Bruce's 2024 top-up outside the C$1.53 billion to C$2.04 billion band, the reconciliation's central bound fails. If the EB-2025-0297 Decision and Order sets payment amounts materially below the settlement's, branch B overstates GA-OPG and the drift-reversal conclusion falls with it. If actual Total GA for 2027 lands outside the grid's C$6.79 billion to C$11.58 billion range on an unchanged perimeter, the pool machinery is mis-specified in a way the backtest did not reveal. If a published generation-by-fuel outlook shows the mix shifting far enough that the Section 7.2 bounds bind, the constant-mix forecast understates its own error. This is a retrospective analysis; there was no pre-registered plan.

12 Conclusion

Ontario's Global Adjustment collected C$94.48 billion over the ten years to September 2026, twice the wholesale value of the energy it supplements, and it did so through a decomposition that is closed, published and verifiable at pool level. At facility level the public record supports less: prices for 99.03 percent of generation, totals for a minority of the fleet, and an explained share of 43.5 percent of 2024 GA, with the balance named and its largest block bounded. The contract book's own dates say the charge's priced core decays through the 2030s, with at least C$716.2 million per year of quantified top-up expiring between 2029 and 2039. Whether the total follows that decay depends on the Ontario Energy Board's EB-2025-0297 decision: on current regulated economics Total GA falls in every scenario cell to between C$5.38 billion and C$9.77 billion by 2035, and on the settlement path it rises in every cell, peaking near C$11.33 billion on the central path in 2031, while the Class B rate falls under both because demand grows faster than either numerator. The backtest's lesson governs how all of this should be read: the machinery errs by 3.1 percent in a quiet year and by 23.5 percent across a perimeter change, so the forecast is a statement about the contract book and the regulatory branch, conditional on the perimeter holding. The perimeter has not held for long in the past.

Acknowledgments

This paper was produced by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data. The underlying datasets are maintained in the Kilambi Research IESO data estate and are listed with their vault keys and hashes in Appendix A. This is a retrospective analysis; there was no pre-registered plan.

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A Data appendix

A.1 Series definitions

Total GA is the IESO's published monthly Global Adjustment total, in Canadian dollars, as assembled in the GA actual series (Section 4.1). **GA-OPA**, **GA-OPG** and **GA-OEFC-NUG** are its three published components; the decomposition is closed in every month from 2008, the first year components are published. **Class B rate** is the Class B amount divided by Class B consumption, both as published in the components workbook (2015 to 2025) or the monthly notices (2026); the Class A amount share is Class A dollars over Total GA. **Market leg** is a facility's output-weighted average market price over a period (HOEP before May 2025; the equal-weighted mean of its delivery-point nodal prices after 23 July 2026). **Contract total** is a facility's all-in price where the public record states one: a register schedule price for tariff facilities (base rate, indexation not applied), the GA-implied combined total for the OPG prescribed fleets, and the documented bounds only for Bruce. **Top-up** is contract total less market leg. **Explained share** is the facility-modelled top-up sum (the OPG combined row plus the schedule-priced subset) over actual Total GA. **Commodity value** in the ten-year comparison is the estate's compilation from IESO settlement publications of HOEP times consumption, held in the same data estate and cross-validated against the assembled series to C$0.1 million per year.

Table 8: Total GA and components by year, 2005 to 2026 (C$ billions)

Year	Total GA	GA-OPA	GA-OPG	GA-OEFC-NUG
2005	-1.15	—	—	—
2006	0.65	—	—	—
2007	0.60	—	—	—
2008	0.90	0.61	-0.19	0.48
2009	4.22	1.71	1.55	0.96
2010	3.85	2.00	0.89	0.96
2011	5.31	2.88	1.35	1.08
2012	6.46	3.52	1.85	1.09
2013	7.73	4.84	1.75	1.13
2014	7.04	4.88	1.39	0.76
2015	9.96	6.67	2.57	0.72
2016	12.33	7.97	3.52	0.84
2017	11.85	8.93	2.65	0.27
2018	11.20	8.17	2.89	0.14
2019	12.98	8.76	4.11	0.12
2020	13.96	9.22	4.64	0.10
2021	8.55	5.15	3.34	0.06
2022	6.23	4.05	2.15	0.03
2023	8.70	5.20	3.48	0.03
2024	8.43	5.45	2.95	0.03
2025	5.27	3.59	1.66	0.02
2026 (Jan–Sep)	4.20	—	—	—

Notes: Components are published from 2008. The 2005 total is negative: GA was a credit year. 2026 covers January to September actuals; the components and the class split for 2026 are published only in the components workbook's next annual edition and are not held. *Source:* GA actual series (Observed, A).

A.2 Historical series

A.3 The EB-2025-0297 settlement amounts

A.4 Provenance

Table 9: EB-2025-0297 settlement-adjusted amounts used in forecast branch B

Year	Weighted average payment amount (C$/MWh)	Total regulated payments (C$ B)	Production forecast (TWh)
2027	102.79	5.36	52.2
2028	104.39	6.32	60.5
2029	110.49	6.49	58.7
2030	112.30	6.86	61.1
2031	124.46	8.08	64.9

Notes: The weighted average payment amount approved for 2026, the base the settlement replaces, is C$77.75/MWh. The settlement is comprehensive except for the C$170.2M Capacity Refurbishment Variance Account debit balance (the outstanding issue), which is excluded from the settled riders, and the amounts remain subject to an update for the Ontario Energy Board's prevailing return on equity at the date of the final order. No Decision and Order had been issued at the build date. *Source:* Settlement Proposal, Exhibit O-1-1, filed 5 October 2026, Draft Payment Amounts Order, Appendix B, Table 5 (Observed, A). Total payments and production combine the nuclear (OPG plus DNNP LP) and regulated hydroelectric fleets.

Table 10: Vault provenance of the paper's data files

Dataset	Vault key (Backblaze B2, bucket ieso-data)	SHA-256 prefix	Registry entry
GA actual series	`raw_contracts/register/ga_actual_series.csv`	0ae026d0	`contracts-ga-actual-series`
Contract register	`raw_contracts/register/generator_contract_prices.csv`	676b0214	`ieso-contracts register`
Derived market rates	`raw_contracts/register/generator_derived_market_rates.csv`	b00ccd2d	`ieso-derived-market-rates`
Fuel price panel	`raw_contracts/register/fuel_price_panel.csv`	cd75fd7e	`contracts-fuel-price-panel`
Roll-off schedule	`raw_contracts/register/contract_rolloff_schedule.csv`	b13a0914	`contracts-rolloff`
Replacement prices	`raw_contracts/register/replacement_price_reference.csv`	39d825a4	`contracts-rolloff`
APO demand extraction	`raw_planning/APO/apo2026_demand_forecast_extracted.csv`	e1d91a21	`www-apo2026-demand-extract`
EB-2025-0297 extraction	`raw_contracts/register/eb-2025-0297_extraction.md`	89848bbd	`contracts-eb-2025-0297`
GA forecast grid	`raw_contracts/register/ga_forecast_v1.csv`	6da92708	`contracts-ga-forecast`

Notes: SHA-256 prefixes are the first eight characters of each file's hash at the build date; the derived market rates hash is the standing updater's current version (the table is re-derived daily and only open-period rows change). Copies of the files as used, with full hashes, are held with this paper's build materials. *Source:* Estate pull registry, canonical copy at B2 `_state/pull_registry.json`.